

Deep learning for seismic event detection of earthquake aftershocks

Lijun Zhu, Zhigang Peng, and James McClellan
Georgia Institute of Technology

Abstract—We design a system based on a convolutional neural network (CNN) for event detection and event time picking, and apply it to continuous recordings during aftershocks of the 2008 M_w 7.9 Wenchuan Earthquake. The system detects 97% of the labeled seismic phases in a standard catalog. Off-line training of the CNN remains successful at 95% accuracy after reducing the training set size by a factor of ten to a few thousand training samples. When applied to a continuous aftershock dataset, the CNN-based method achieves accurate detection of manually labeled phases, precise arrival times of picked phases, as well as discovering many weak events not listed in the given seismic catalog. On-line processing of a 31-day recording takes less than 12 hours on a single GPU.

The increasing volume of seismic data from long-term continuous monitoring requires faster and more reliable phase detection and picking algorithms. Seismic event detection methods based on statistical signal processing must make certain assumptions about the observed seismograms and require careful parameter tweaking to achieve good results on different datasets. Deep learning, with a large number of parameters automatically trained on labeled data and facilitated by the parallel computation power of modern graphics processing units (GPUs), has demonstrated accuracy on speech, image and video applications. Thus, we cast the seismic (P and S-wave) phase detection process as a multi-class classification problem and propose a CNN-based Phase-Identification Classifier (CPIC) for single-station multi-channel seismic waveforms. The CNN weights are obtained via supervised learning based on a catalog containing tens of thousands of human-labeled phase and non-phase samples used in a recent competition for detecting aftershocks of the M_w 7.9 Wenchuan Earthquake [1].

I. DATA

Many recent CNN studies for seismic rely on an exceptionally rich training dataset of labeled samples [2] to achieve good accuracy and robustness against noise. However, we design CPIC and study its performance on a relatively small training set prior to applying it

JM is supported by the John and Marilu McCarty Chair; ZP is supported by National Science Foundation grant EAR-1818611. LZ is supported by the McCarty Chair and USC/SEEC grant 18165.

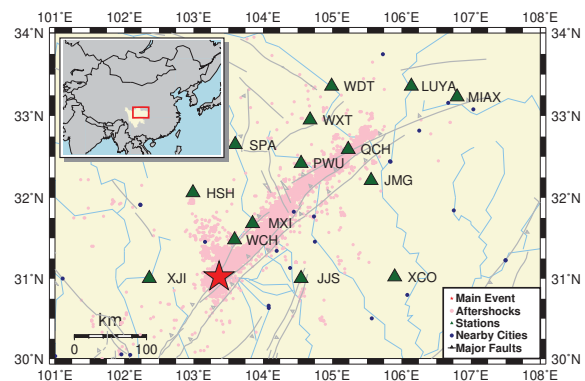


Fig. 1: Map showing the study region in Sichuan, China along the aftershock zone of the 2008 M_w 7.9 Wenchuan earthquake (red star). The 9,648 manually picked aftershocks are marked as pink dots. Green triangles mark the 14 permanent stations used in this study. Gray and blue thin lines mark active faults and rivers in this region.

on a large volume of unlabeled data. This is a typical scenario when analyzing seismic data following a major earthquake: strong aftershocks can be easily picked by existing algorithms or analysts, but the real targets are the numerous small events that might be missed by traditional methods [3], [4].

For CPIC minimal pre-processing is applied on the seismogram: only soft-clipping with a logistic function is used to normalize the large dynamic range of the input waveforms. As will be shown in Fig. 5, such pre-processing contributes to CPIC's fast convergence as well as higher accuracy. Notably, no bandpass filtering is applied to seismic waveforms in pre-processing step.

1) *Study region*: We utilized the aftershock dataset of 2008/05/12 M_w 7.9 Wenchuan earthquake that was made available during a recent competition for identifying seismic phases [1]. The mainshock occurred on the eastern margin of Tibetan Plateau (Fig. 1), and ruptured the central and northern section of the Longmenshan fault zone [5]. Numerous aftershocks occurred following the mainshock, but many of them are still missing in any published earthquake catalogs [6]. The aftershock dataset

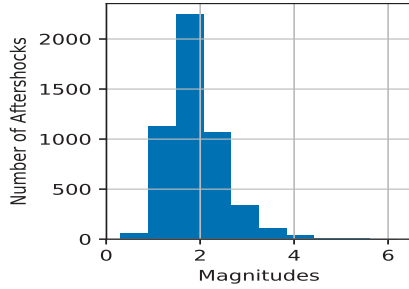


Fig. 2: Magnitude M_w distribution of catalog events.

includes one-month of continuous data recorded by 14 permanent stations in August 2008, about three months after the Wenchuan mainshock. There are 30,327 phases manually picked on these stations: 15,281 P phases and 15,046 S phases. Stations that are near the rupture zones and aftershocks (e.g., PWU, MXI, WXT, JMG and QCH) had most of the picked phases while the distant stations (e.g., XCO, MIAX, LUYA, and SPA) have many fewer labeled phases with time picks.

2) *Labeled dataset*: In this work, CPIC is trained on a dataset of labeled seismic frames chosen based on the given phase labels and their time picks. For each time pick, we define a 20-sec long window starting 5-sec before and ending 15-sec after as one frame of a seismic phase (Fig. 3). We deliberately choose a long time window so that there is a high likelihood that a P-wave frame contains some S-wave at its end, and that S-wave frames contain some P-wave coda at the beginning. This window definition implicitly embeds the normal sequential relationship between P and S wave phases in the labeled dataset itself. Adding noise-only frames, which are not included in the original labeled dataset helps to improve the trained CPIC's performance against noisy seismograms. For labeled noise frames, we assume that quiet regions exist about 60 sec after an S-wave phase or 60 sec before a P-wave phase and generate 30,000 noise-only frames. Since these noise frames are not verified by an analyst, it is possible that they may include small aftershocks not listed in the catalog. In the end, we obtain a dataset with roughly 60,000 labeled frames, for which P-wave, S-wave, or noise labels have been assigned.

3) *Continuous dataset*: Once CPIC is trained on the labeled dataset, the phase detector and arrival-time picker are then tested on the entire one-month continuous waveforms starting on 08/01/2008 00:00:00 Beijing Time. Due to challenging acquisition conditions in the study area, there are some gaps in the continuous recording. These are filled with zeros to keep the overall dataset consistent while avoiding artificial detections.

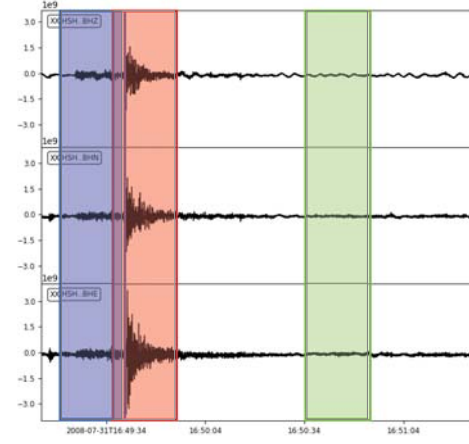


Fig. 3: An example of three-component seismogram recorded at station HSH. We use 20-sec time frames extracted for both P (blue) and S (red) phases. Noise (green) frames are cut 60 sec before P phases, or 60 sec after S phases. Sampling rate is 100 Hz.

II. METHOD

We developed the CPIC processing pipeline summarized in Fig. 4. An offline training process is used

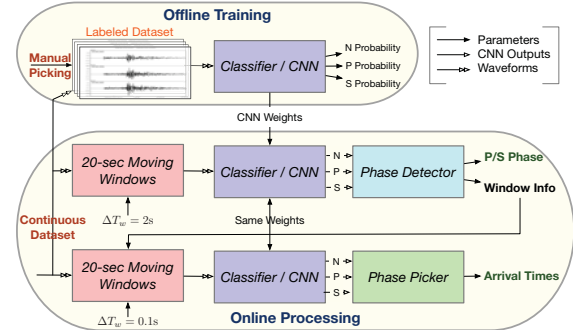


Fig. 4: Flow chart of CPIC processing pipeline. Inputs are labeled in red and outputs in green. The 20-sec moving windows are overlapped with the offset ΔT_w .

to optimize the parameters of the CNN classifier with iterative optimization over the labeled dataset of 60,000 frames.

1) *CNN Details*: Shown in Table I, the CNN inputs are 3-C seismograms of labeled 20-sec frames, sampled at 100 Hz. The CNN network is adapted from [7], which uses the same size kernel at each convolutional layer and obtains increasing accuracy as the network goes deeper. Following [8], we modify the first two layers with larger kernels to increase the efficiency of extracting large-scale features at the beginning.

The three classifier outputs are probabilities of each frame containing noise, P phase, or S phase. The *Softmax* function is used to normalize the probabilities in the output layer. Hence, the *Softmax* classifier minimizes the cross-entropy between the estimated class probabilities (q) and the true distribution, where all probability mass is on the correct class, e.g., $p = (0, 1, 0)$ for a labeled P phase frame. Between each layer, a rectified linear unit (ReLU) activation function introduces nonlinearity into the model. The data size is reduced at each layer using max-pooling, and optimization of the training problem is performed using Adam [9]. Batch normalization is also integrated to further stabilize the training process. A fully-connected (FC-64) layer follows the convolutional layers to perform linear classification from a 64-element feature vector to three output classes.

TABLE I: Proposed CNN architecture with 11 convolutional layers and one fully-connected layer with three outputs that give the probability of a seismogram frame being noise, P, or S phase.

input shape	kernel size	max pooling	#params
2000×3	$5 \times 1 \times 3$	$2000 \rightarrow 1000$	240
1000×16	$5 \times 1 \times 16$	$1000 \rightarrow 500$	2560
500×32	$3 \times 1 \times 32$	$500 \rightarrow 250$	6144
250×64	$3 \times 1 \times 64$	$250 \rightarrow 128$	12288
128×64	$3 \times 1 \times 64$	$128 \rightarrow 64$	12288
64×64	$3 \times 1 \times 64$	$64 \rightarrow 32$	12288
32×64	$3 \times 1 \times 64$	$32 \rightarrow 16$	12288
16×64	$3 \times 1 \times 64$	$16 \rightarrow 8$	12288
8×64	$3 \times 1 \times 64$	$8 \rightarrow 4$	12288
4×64	$3 \times 1 \times 64$	$4 \rightarrow 2$	12288
2×64	$3 \times 1 \times 64$	$2 \rightarrow 1$	12288
1×64	FC-64		192
1×3	softmax (normalize to 1)		

To accommodate the modest training set size, the proposed CNN uses only one convolution layer between each max-pooling layer. There are 107,440 parameters in the CNN+FC for the 20-sec window case, but this number could be reduced with a shorter window length.

2) *Continuous Phase Detection and Picking*: The trained classifier, which is optimized to give P, S and noise (N) probabilities for individual 20-sec frames, also supports continuous online processing, which can be accomplished in two steps:

Phase detection: identify frames with a seismic event

Phase picking: determine event time within a frame

For a single 20-sec time frame, *phase detection* uses the CNN output probabilities of a frame being a specific seismic (P or S) phase, or a noise frame to make a decision. When either the P or S phase probability is greater than 0.5, a seismic phase detection is made;

otherwise, the frame is declared noise. During online processing, the continuous waveforms are broken into overlapping 20-sec frames. With 90% overlap the offset is $\Delta T_w = 2$ sec which produces probabilities continuously for detecting frames with P or S phases, using the 0.5 threshold. Neighboring detected frames from the same event are merged into a longer window.

The *phase picking* step is an estimation problem where a value (arrival time) is obtained from the detected (merged) window. In order to achieve sufficient time resolution, the CNN must be rerun on a finer grid, e.g., $\Delta T_w \approx 0.1$ sec, i.e., 99.5% overlap. The CNN output probabilities are used to produce two time sequences, called characteristic functions (CFs). These are ratios, $\log_{10}(P/N)$ and $\log_{10}(S/N)$, which represent the likelihood of a time frame being a P or S phase. Since the CFs are ratios with respect to noise, using a fixed threshold will effectively adapt to changing noise levels.

The CNN outputs not only give the probabilities for a frame, but also the probability associated with the time pick being at 5 sec from the frame beginning. Thus, the finely sampled CF time sequences give an estimator for the arrival time which is done by finding peaks.

III. CNN CLASSIFIER TRAINING & TESTING

To systematically verify the accuracy and stability of the proposed networks, the available 60,276 samples are then randomly split into two parts: two thirds for training (40,184) and one third for testing (20,092). This random split is repeated several times to make sure the networks are stable and the final results are reliable. Our choice of the split ratio is arbitrary, but was made to have as many training samples as possible while keeping the testing set large enough for reasonable validation. The training set is repeatedly fed into the neural network to minimize the error in batches of 100 frames. After the CNN sees every frame in the entire training dataset once, we have finished one epoch of training. Multiple epochs are taken to fully train the networks into a stable state.

During training the CNN is tested on 20,000 events to evaluate the accuracy after every epoch. Figure 5 shows the training loss (red) and testing accuracy (blue) with respect to the number of epochs. The proposed network (solid lines) reaches 97% accuracy after five epochs and becomes stable even though the training losses keep going down. We observed that the training process usually converges to its final accuracy in less than ten epochs. The use of soft clipping of the raw seismograms is essential. Without soft clipping (dashed lines), the accuracy of the networks slowly increases but with large oscillation centered around 85% accuracy even though the training losses also goes down. Finally, applying the trained 11-layer CNN only takes about

0.4 msec per frame on average, when feeding inputs in batches of 100 frames.

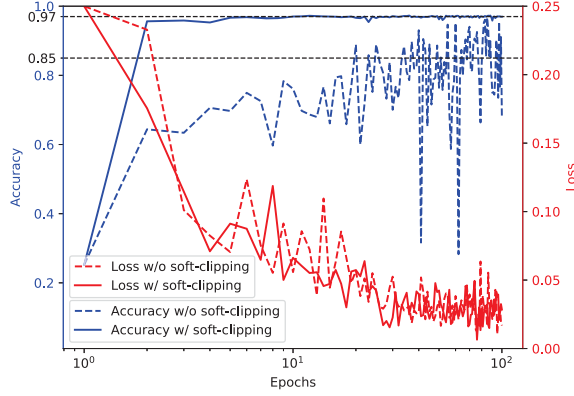


Fig. 5: Effect of soft-clipping: training error (red) and testing accuracy (blue) with respect to number of epochs. Solid curves show the soft-clipped cases while dashed curves show the non-soft-clipped cases.

1) *Vary Training Set Size*: With different splitting ratios we explored the minimum required training dataset size (see Fig. II). When going from 43,969 training samples to 3,657, the average detection accuracy drops from 97% to only 95%. This suggests that a much smaller catalog of manually picked aftershock events would be sufficient to achieve good detection accuracy, which would reduce the amount of manual labeling needed to a reasonable level for many scenarios.

TABLE II: Testing accuracy for different training dataset size averaged over 40 Monte-Carlo runs.

Training Samples	Testing Samples	μ accuracy	σ accuracy
415	59,861	0.862	0.023
773	59,503	0.901	0.015
1,439	58,837	0.925	0.013
2,680	57,596	0.943	0.005
3,657	56,619	0.949	0.006
4,991	55,285	0.953	0.006
6,810	53,466	0.957	0.004
12,681	47,595	0.961	0.008
23,613	36,663	0.965	0.005
43,969	16,307	0.972	0.010

2) *Testing CNN Classifier*: When the trained CNN classifier-detector is run on the 20,000 labeled frames reserved for testing, we get a detailed breakdown of the true and false decisions for the three classes (P, S and N). Table III shows the resulting confusion matrix. Each row shows the number of detections made for one class label, while the columns are the CPIC labels. In the P-wave row 4,897 P phases are detected correctly, 60 are detected as noise, and 7 as S-wave events, which translates to a *recall* of $4897/4964 = 0.987$; the P phase

precision is $4897/5074 = 0.965$. For the S phase, recall is 0.978 and precision 0.957.

TABLE III: Confusion matrix for detection of seismic phases on one-month Wenchuan aftershock labeled dataset with 40,000 training and 20,000 testing samples.

CATALOG	DETECTOR			Total
	Noise	P-wave	S-wave	
Noise	10,064	165	212	10,441
P-wave	60	4,897	7	4,964
S-wave	105	12	4,816	4,933
Total	10,229	5,074	5,035	20,338

IV. CONTINUOUS PROCESSING RESULTS

1) *Event Detection and Time Picking*: Figure 6c shows the *phase detection* process operating on a 300-sec continuous waveform. The case where $\Delta T_w = 2$ sec shown in Fig. 6c is the best offset for this example to find both a strong event (near 100 sec) and weak ones (near 50 sec and 250 sec), while not introducing too many short duration spikes which are likely to be false positives.

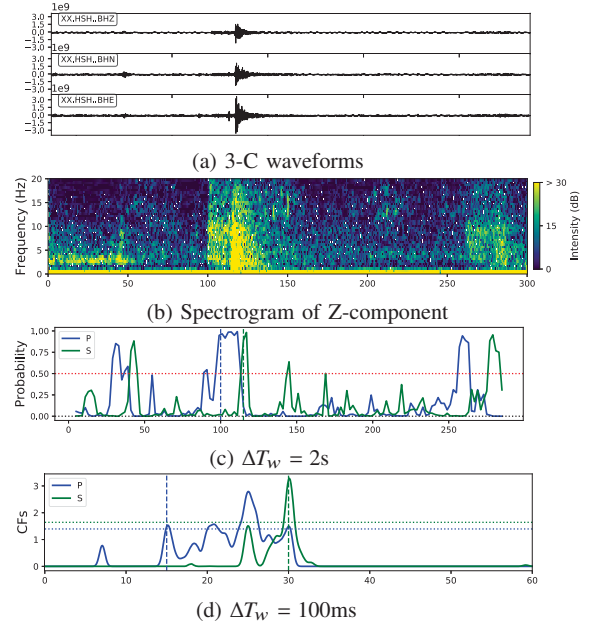


Fig. 6: Phase detection and picking on (a) 300-sec continuous 3-C waveforms, (b) spectrogram, (c) probabilities for P and S phases; horizontal dotted red line is detection threshold at 0.5, (d) characteristic functions (P blue, S green) for 60 sec window starting at 85 sec with $\Delta T_w = 100$ ms. The catalog picks of P (blue) and S (green) phases are marked by dashed vertical lines.

The picking results (with a 100 ms moving window) of P and S phases are shown together in Fig. 6d. Arrival

times are picked at the *first valid peak* of both CFs. To avoid small peaks due to noise, we use an adaptive threshold at half the global peak of each smoothed CF. For example, the P phase global peak is at (25.0 s, 2.79), so the $\log_{10}$ threshold is set to 1.40 which is $10^{1.4} \approx 25$ times smaller than the global peak. This removes the peak at 8 sec which is less than 1.0 (or more than 63 times less than the global peak). The global peak for the S phase is at (30.1 s, 3.28) which gives a threshold at 1.64. Although the S-peak at 25 sec is above this threshold, it is not recognized by the detector as S phase since the P phase has a higher probability. So, the first valid peaks for P and S phases are at (15.0 s, 1.533) and (30.1 s, 3.28). In this case, these match the catalog picks closely. The complete statistics of the error between picked and catalog arrival times are shown in Fig. 7.

The next test is to run the 2-sec offset detector over the entire 31-day continuous dataset. In addition to the catalog events, the CPIC detector found 119,066 candidate time windows not listed in the catalog. To investigate these additional phase detections, we built a matched-filter (MF) enhanced catalog for one day (8/30/2008) following the procedure used by [10] during which 968 phases are listed in the catalog. This MF catalog expands the original 968 phases into 12,200 phases. For that day, CPIC detects 4,123 seismic phase frames among which 2,892 (70%) contain a phase in the MF catalog. Further studies are needed to check whether the remaining 30% might correspond to actual events that are not similar to existing templates.

2) *Accurate Phase Picking*: To perform phase picking, the merged windows from the detector must be run through the CNN again but with a much smaller offset. Figure 6d shows that the CNN classifier responds differently to P and S phase arrivals: the true P arrival is closer to the rising edge of its CF, which is confirmed by the spectrogram of Z-component in Fig. 6b, while the S arrival is closer to its peak.

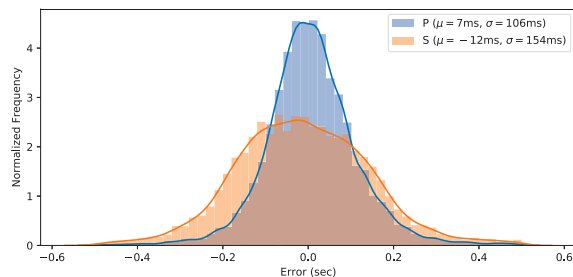


Fig. 7: Histograms (normalized to unit area) of P and S phase time picking errors.

For the detected frames with a catalog phase, we can evaluate the error between the CNN phase picks and the manual picks to measure the performance of the P

and S phase pickers separately. Picks that are more than 0.5 sec away from the manual picks are deemed outliers, and we find 1.26% of the P picks and 4.62% of the S picks. The errors for the non-outliers are displayed in the histograms of Fig. 7. Both phase pickers have very small bias ($\mu = 7$ ms for P, and $\mu = -12$ ms for S), and the standard deviation of the P phase picker ($\sigma = 106$ ms) is smaller than for the S phase picker ($\sigma = 154$ ms). This is expected since picking S phase arrivals is more difficult in general, and its manual picks are also less reliable.

CPIC tests are run on an Nvidia GTX 1080 Ti GPU with 3,584 CUDA cores and 11 GB memory. Online processing of one 20-sec frame by the trained CNN classifier takes less than 0.3 ms on average when feeding the input as 1000 frames per batch to exploit the maximum GPU memory size. Running time for the event detector alone on the entire 31-day continuous 3-C waveforms from 14 stations is less than two hours. Finally, CPIC takes around 12 hours to pick all 30,000 labeled catalog phases within the 31-day dataset.

REFERENCES

- [1] L. Fang, Z. Wu, and K. Song, "Seismolympics," *Seismological Research Letters*, vol. 88, no. 6, p. 1429, 2017.
- [2] W. Zhu and G. C. Beroza, "Phasenet: A deep-neural-network-based seismic arrival time picking method," *arXiv preprint arXiv:1803.03211*, 2018.
- [3] Y. Y. Kagan, "Short-term properties of earthquake catalogs and models of earthquake source," *Bulletin of the Seismological Society of America*, vol. 94, no. 4, pp. 1207–1228, 2004.
- [4] Z. Peng, J. E. Vidale, and H. Houston, "Anomalous early aftershock decay rate of the 2004 M_w 6.0 Parkfield, California, earthquake," *Geophysical Research Letters*, vol. 33, no. 17, 2006.
- [5] G. Feng, E. A. Hetland, X. Ding, Z. Li, and L. Zhang, "Coseismic fault slip of the 2008 M_w 7.9 Wenchuan earthquake estimated from InSAR and GPS measurements," *Geophysical Research Letters*, vol. 37, no. 1, L01302, 2010.
- [6] X. Yin, J.-H. Chen, Z. Peng, X. Meng, Q. Liu, B. Guo, and S. Cheng Li, "Evolution and distribution of the early aftershocks following the 2008 M_w 7.9 Wenchuan earthquake in Sichuan, China," *J. Geophys. Res.: Solid Earth*, 2018.
- [7] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *CoRR*, vol. abs/1409.1556, 2014.
- [8] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2015.
- [9] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [10] X. Meng, Z. Peng, and J. L. Hardebeck, "Seismicity around Parkfield correlates with static shear stress changes following the 2003 M_w 6.5 San Simeon earthquake," *J. Geophys. Res.: Solid Earth*, vol. 118, no. 7, pp. 3576–3591, 2013.